

Decision decisions

In this project we've used a combination of design patterns. The singleton pattern was used to create the main classes and actors of the project, these include the Device, Headset and PC. Given the persistent aspect of each actor in the scenarios, it made sense to create a single instance that persists throughout the lifetime of the application. The Observer pattern is very prominent in the Qt framework, the UI of the application relies on the signals to be handled by slot functions connected to them by the MainWindow. As such, we attempt to leverage this design pattern where possible to delegate the concurrency of certain operations, where QThreads might have been difficult to manage.

Use cases

Use Case #1: Normal operation of treatment with Neureset device

Name: Normal operation of treatment with a Neureset device

Primary Actor(s): Device operator (user)

Stakeholders and interests:

- Patient
- Operator
- Therapist
- Health institution

Pre-condition(s):

- Patient is present and wearing EEG headset
- The device has power and is turned on
- The device is connected to the EEG headset
- The device has sufficient power to deliver the full treatment.

Success guarantee(s): User successfully completes a therapy session

Main success scenario:

1. Navigate to the 'new session' menu option and select it.
2. The session information is displayed to the user including the approximate time remaining and session progress.
3. Record the frequencies at different wavelengths at every node on the headset over one minute, the baseline frequency and amplitude of the wavelengths are used to calculate the dominant frequency.
4. The frequencies and amplitudes are recorded in the session's log.
5. The waveform is displayed to the user.
6. A treatment frequency is applied to each site that is equal to the dominant frequency of that site plus an offset. The calculated frequency is applied every 1/16 seconds over one second.
7. The green light flashes for one second while the treatment is being delivered.
8. The treatment frequency used in the treatments is stored in the session's log.
9. The device repeats steps 6-8, increasing the offset applied to the dominant frequency by 5 Hz. The treatment is finished after the offset added to the dominant frequency equal to 20 Hz has been administered.
10. Repeat step 4, storing the frequencies after the session has been administered.

11. The session log is saved to the device, and ready to transfer to the PC.

Extensions:

1a. If the headset isn't connected before a therapy session, a new session will not start. A blue light indicates the headset and the device are connected

2-10. Headset connection is lost

2-10b1. Pauses the current session

2-10b2. A red light flashes and the device starts beeping

2-10b3. If contact is reestablished, the session resumes from where it was paused, and the warnings from 2b2 are stopped.

2b4. If contact is not reestablished after 5 mins, turn off the device and erase the session log.

3-10a. The user can pause the active therapy session voluntarily.

3-10a1. Pause the session, save the current state

3-10a2. The session resumes from the same state where it was paused

3-10a3. If the session is not resumed after 5 minutes, turn off the device and erase the session

*a. At any moment the device detects low battery, redirect to Battery low response of the device
Use Case #3

Use Case #2: Update the current date and time

Name: User wants to update the current date and time

Primary Actor(s): User

Pre-condition(s):

- The device has power and is turned

Success guarantee(s): The date and time are updated based on user input

Main success scenario:

1. Navigate to the date and time setting menu option
2. Change the date and time to the correct time
3. Save changes

Use Case #3: Battery low response of the device

Name: Battery low response of the device

Primary Actor(s): User

Pre-condition(s):

- The user of the device has another battery to replace the current battery

Success guarantee(s): The device battery has been changed and has enough power to continue normal operations

Main success scenario:

1. A low battery icon is displayed in the corner of the screen.
2. A popup notification is displayed warning the user about the low battery.
3. The power button on the device is pressed, turning off the device.
4. The operator replaces the device battery by sliding the cover off the battery compartment on the back of the device, inserting a new battery, and sliding the cover back into place.

5. The operator presses the power button to turn on the device.
6. The low battery notification disappears if the battery has sufficient charge.

Extensions:

- 1a. Low battery should last enough to finish an ongoing treatment
- 1b. Users should not be able to start a new session while the battery is low
- 2a. User can dismiss the notification
- 3a. If the device battery is completely depleted, the device shuts down automatically.

Use Case #4: Therapy history viewing with PC

Name: Therapy history viewing with PC

Primary Actor(s): Patient

Stakeholders and interests:

- Patient
- Operator
- Therapist
- Health institution

Pre-condition(s): The Neurofeedback device has power and is turned on.

Success guarantee(s): Successfully displaying therapy history data on a PC for interpretation.

Main success scenario:

1. Connect the device to a PC
2. The user presses the “session log history” option on the device menu and can scroll through the list of sessions displayed on the device screen.
3. Select “upload session logs” to upload all current existing logs to PC without duplicates.
4. The device uploads the entire session data to the connected PC.
5. The PC stores and displays a list of the logs uploaded
6. The user can select a session log from the list to view more details about the session
7. Session details include session number and date, before and after baselines per site and EEG waveforms recorded

Extensions:

- 1a. Unable to connect the device to a PC.
 - 1a.1 Turn off the device and turn it on. Try connecting to the PC again.
- 2a. The device display does not update after pressing the “session log history” option.
 - 2a.1 Turn off the device and turn it on. Try pressing the “session log history” option again.
- 3a. The desired log session does not appear on the device screen.
 - 3a.1 Wait a few minutes for data of the latest session to be saved in the session log history
 - 3a.2 Turn off the device and turn it on again. Repeat step 2.
- 1-3a. The device runs out of battery while it is connected to the PC.
 - 1-3a.1 Refer to Battery low response of the device Use Case #3.
- 4b. The device runs out of battery during session log upload.
 - 4b.1 Refer to Battery low response of the device Use Case #3 and repeat from step 1.
- 5a. The uploaded data is corrupted and cannot be displayed on the PC screen.
 - 5a.1 Retry uploading the same session log to PC.